

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 30-08-2026

1. TITLE

Develop a College Information Portal Using HTML Text Formatting, Tables, Forms, and HTML5 Semantic and Multimedia Elements

2. AIM

To design and develop a static College Information Portal web page using HTML that demonstrates text formatting tags, ordered and unordered lists, hyperlinks, tables, images, a multi-field registration form, and HTML5 semantic and multimedia elements.

3. OBJECTIVE

The objectives of this experiment are:

- To use basic HTML text formatting tags such as bold, italic, and underline to present content.
 - To apply subscript and superscript tags to represent chemical formulae and mathematical expressions.
 - To organize content using ordered and unordered lists.
 - To create hyperlinks to internal sections and external websites.
 - To display tabular data using table, tr, th, and td elements with borders, padding, and spacing.
 - To embed images with width, height, alt text, and a fallback source using the onerror attribute.
 - To build a registration form using text, email, password, radio, checkbox, select, and textarea input types.
 - To apply HTML5 semantic elements such as header, nav, section, article, and footer to structure a page meaningfully.
 - To embed multimedia content using the audio, video, and iframe elements.
 - To use HTML5 input types such as email, date, number, and range for modern interactive form controls.
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4. THEORY

HTML Text Formatting — the `<b>`, `<i>`, and `<u>` tags render bold, italic, and underlined text respectively, while `<sub>` and `<sup>` lower or raise text for chemical formulae and exponents.

HTML Lists — the `<ol>` element numbers its `<li>` items automatically, while the `<ul>` element marks each `<li>` with a bullet; both are used to organise related pieces of content.

HTML Tables — the `<table>` element together with `<tr>`, `<th>`, and `<td>` arranges data into rows and columns; the `border`, `cellpadding`, and `cellspacing` attributes control the table's visible grid and internal spacing.

HTML Images — the `<img>` element displays a picture using its `src` attribute, with `width` and `height` fixing its display size and `alt` providing descriptive fallback text; the `onerror` attribute can load a backup image if the primary source fails.

HTML Forms — the `<form>` element groups related input controls; `label for="..."` ties descriptive text to a specific field, and different input type values (text, email, password, radio, checkbox) capture different kinds of data, while `<select>` and `<textarea>` provide dropdown and multi-line input respectively.

HTML5 Semantic Elements — `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` describe the role of each part of the page, which helps browsers, assistive technology, and search engines interpret the page structure correctly.

HTML5 Multimedia — the `<audio>` and `<video>` elements embed playable media directly in the page using the `controls` attribute, and an `<iframe>` embeds an external page such as a video player inside the current document.

HTML5 Input Types — input types such as email, date, number, and range give the browser built-in validation and specialised controls (a calendar picker for dates, a stepper for numbers, and a draggable slider for a numeric range) without any additional scripting.

5. ALGORITHM / PROCEDURE

- Create a single file named `index.html` and set up the standard HTML5 boilerplate with a `charset` meta tag, a `viewport` meta tag, and a page title.
- In the header, add a banner image with a `width` attribute and an `onerror` handler that swaps in a fallback logo if the primary image fails to load.
- Under "Part A - Basic HTML", add a paragraph describing the college using bold, italic, and underline tags, followed by a second paragraph demonstrating subscript and superscript.
- Add an ordered list of courses offered and an unordered list of campus facilities.
- Add a paragraph of anchor tags linking to the official college website and an external search engine, each opening in a new tab.
- Under "Part B - Tables and Images", build a bordered table with a header row (Roll No, Name, Branch, Email) and three data rows of sample student records.
- Add two images below the table — the official logo and a campus building photo — each with fixed `width` and `height` attributes.

- Under "Part C - Student Registration Form", build a form with labelled text, email, and password inputs, a set of radio buttons for gender, a set of checkboxes for hobbies, a select dropdown for branch, and a textarea for comments.
 - Add submit and reset buttons to the form.
 - Under "Part D - HTML5 Features", add a second header containing a heading and a nav with three anchor links to page sections.
 - Add a section with id="intro" explaining semantic layout and an article giving a department overview.
 - Add a section with id="media" containing an audio player, an embedded YouTube iframe, and a video element with a poster image and fallback text.
 - Add a section with id="inputs" with a form containing email, date, number, and range HTML5 input controls.
 - Add a footer with a copyright line and close the page.
 - Open index.html in a browser and verify that each part (text formatting, lists, links, table, images, form, semantic sections, multimedia, and modern inputs) renders and behaves as expected.
 - Record expected vs. actual output for each test case.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>ANITS - College Information Portal</title>
</head>
<body>

  <!-- ===== ANITS BANNER ===== -->
  <header>
    
```

</header>

<h1>College Information Portal</h1>

<!-- ===== PART A ===== -->

<h2>Part A - Basic HTML</h2>

<div>

<h3>About ANITS</h3>

<p>

Anil Neerukonda Institute of Technology and Sciences (ANITS)
is a premier engineering institution that provides
quality education and actively encourages
<i>innovation</i>, <u>research</u>, and practical engineering skills.

</p>

<h3>Text Formatting Examples</h3>

<p>

Bold Text | <i>Italic Text</i> | <u>Underlined Text</u>

Chemical formula: H₂O

Mathematical expression: $x^2 + y^2$

</p>

</div>

<div>

<h3>Courses Offered</h3>

Computer Science & Engineering
Information Technology
CSE (AI & ML)
Electronics & Communication Engg.

<h3>Campus Facilities</h3>

- Central Library & e-Learning Center
- High-Performance Computing Labs
- In-Campus Hostel Facility
- Sports & Athletic Grounds

</div>

<h3>Useful Websites</h3>

<p>

ANITS Official Website |

Google Search

</p>

<hr>

<!-- ===== PART B ===== -->

<h2>Part B - Tables and Images</h2>

<h3>Student Details</h3>

<table border="1" cellpadding="8" cellspacing="0">

<tr>

<th>Roll No</th>

<th>Name</th>

<th>Branch</th>

<th>Email</th>

</tr>

<tr>

<td>A24126552093</td>

<td>GANESH</td>

<td>CSE(AIML)</td>

<td>GANESH@gmail.com</td>

</tr>

<tr>

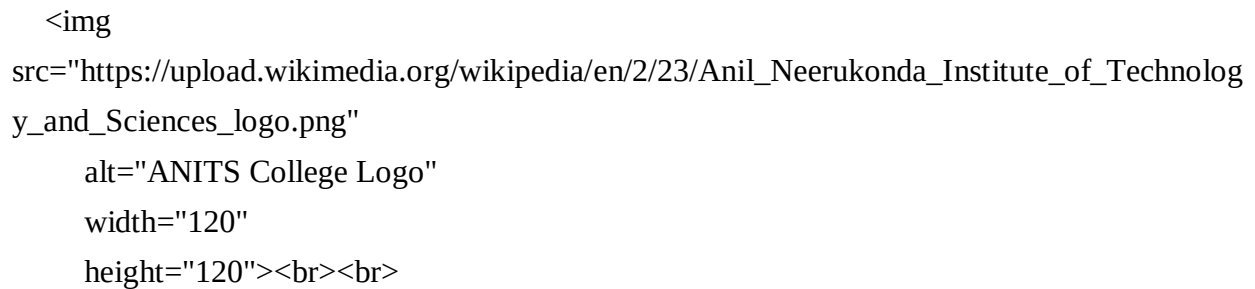
<td>A24126552092</td>

<td>TARUN</td>

CSE	TARUN@gmail.com
A24126552094	PRABHAT
IT	PRABATH@gmail.com

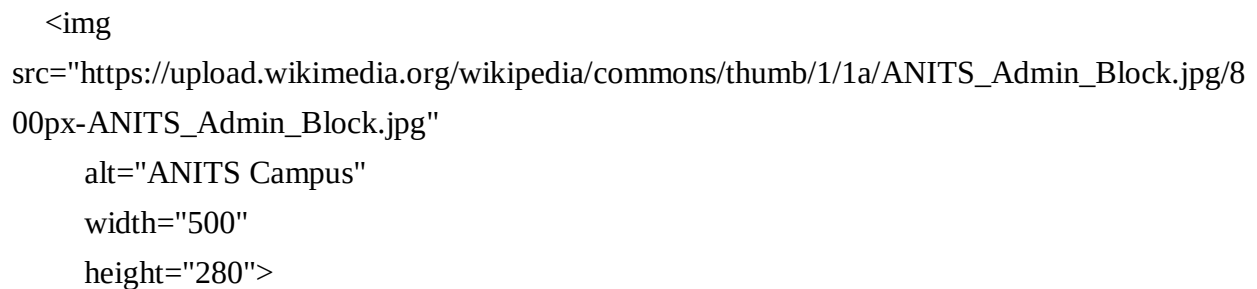
College Brand & Campus

Official Logo:



src="https://upload.wikimedia.org/wikipedia/en/2/23/Anil_Neerukonda_Institute_of_Technology_and_Sciences_logo.png"
alt="ANITS College Logo"
width="120"
height="120">

Campus Building (Admin Block):



src="https://upload.wikimedia.org/wikipedia/commons/thumb/1/1a/ANITS_Admin_Block.jpg/800px-ANITS_Admin_Block.jpg"
alt="ANITS Campus"
width="500"
height="280">

----- PART C -----

Part C - Student Registration Form

<form>

<label for="studentName">Full Name:</label>


```
<input type="text" id="studentName" name="studentName" placeholder="Enter your full name" required>
```

```
<br><br>
```

```
<label for="studentEmail">Email Address:</label><br>
```

```
<input type="email" id="studentEmail" name="studentEmail" placeholder="Enter your email" required>
```

```
<br><br>
```

```
<label for="password">Password:</label><br>
```

```
<input type="password" id="password" name="password" required>
```

```
<br><br>
```

```
<label>Gender:</label><br>
```

```
<input type="radio" id="male" name="gender" value="Male">
```

```
<label for="male">Male</label>
```

```
<input type="radio" id="female" name="gender" value="Female">
```

```
<label for="female">Female</label>
```

```
<input type="radio" id="other" name="gender" value="Other">
```

```
<label for="other">Other</label>
```

```
<br><br>
```

```
<label>Hobbies:</label><br>
```

```
<input type="checkbox" id="reading" name="hobby" value="Reading">
```

```
<label for="reading">Reading</label>
```

```
<input type="checkbox" id="music" name="hobby" value="Music">
```

```
<label for="music">Music</label>
```

```
<input type="checkbox" id="coding" name="hobby" value="Coding">
```

```
<label for="coding">Coding</label>
```

```
<br><br>
```

```
<label for="branch">Select Branch:</label><br>
```

```
<select id="branch" name="branch">
```

```
<option value="">-- Choose Department --</option>
```

```
<option value="cse">CSE</option>
```

```
<option value="it">IT</option>
<option value="csm">CSM (AI & ML)</option>
<option value="ece">ECE</option>
</select>
<br><br>

<label for="comments">Comments / Queries:</label><br>
<textarea id="comments" name="comments" rows="4" cols="50" placeholder="Type your
message here..."></textarea>
<br><br>

<input type="submit" value="Register">
<input type="reset" value="Clear Form">
</form>

<hr>

<!-- ===== PART D ===== -->
<header>
  <h2>Part D - HTML5 Features</h2>
  <nav>
    <a href="#intro">Introduction</a> |
    <a href="#media">Media Elements</a> |
    <a href="#inputs">Modern Inputs</a>
  </nav>
</header>

<section id="intro">
  <h3>HTML5 Semantic Layout</h3>
  <p>
    Semantic elements give meaningful structural context to web documents,
    allowing search engines and browsers to understand headers, navigation bars, content
    sections, and footers easily.
  </p>
</section>
```


<article>

<h3>Department Overview</h3>

<p>

ANITS offers accredited undergraduate B.Tech programs under premier technical disciplines with high placement records.

</p>

</article>

<section id="media">

<h3>Multimedia Showcase</h3>

<p>Audio Player:</p>

<audio controls>

<source src="https://www.soundhelix.com/examples/mp3/SoundHelix-Song-1.mp3" type="audio/mpeg">

Your browser does not support audio playback.

</audio>

<p>ANITS YouTube Campus Tour:</p>

<iframe width="560" height="315"

src="https://www.youtube.com/embed/dQw4w9WgXcQ"

title="ANITS Tour"

frameborder="0"

allowfullscreen>

</iframe>

<p>HTML5 Video Player:</p>

<video width="400" controls

poster="https://upload.wikimedia.org/wikipedia/commons/thumb/1/1a/ANITS_Admin_Block.jpg/800px-ANITS_Admin_Block.jpg">

<source src="https://interactive-examples.mdn.mozilla.net/media/cc0-videos/flower.mp4" type="video/mp4">

Your browser does not support video playback.

</video>

```

</section>

<section id="inputs">
  <h3>HTML5 Interactive Controls</h3>
  <form>
    <label for="email">Email Control:</label>
    <input type="email" id="email" name="email" placeholder="name@domain.com">
    <br><br>
    <label for="date">Joining Date:</label>
    <input type="date" id="date" name="date">
    <br><br>
    <label for="age">Age (17 - 30):</label>
    <input type="number" id="age" name="age" min="17" max="30" value="18">
    <br><br>
    <label for="skill">Skill Proficiency (0 - 10):</label>
    <input type="range" id="skill" name="skill" min="0" max="10" value="5">
  </form>
</section>

<!-- Footer -->
<footer>
  <hr>
  <p>&copy; 2026 ANITS College Information Portal | Department of CSE (AI & ML)</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
, <i>, <u>	Render bold, italic, and underlined text for emphasis within a paragraph.
<sub>, <sup>	Lower or raise text to represent chemical formulae (H ₂ O) and exponents (x ² + y ²).
,	Display the list of courses as numbered items and the list of facilities as bulleted items.

Code	Explanation
<code></code>	Creates a hyperlink that opens the linked page in a new browser tab.
<code><table>, <tr>, <th>, <td></code>	Arrange the student records into a bordered grid of rows and columns.
<code></code>	Displays the banner image and swaps to a fallback source if the first one fails to load.
<code><form>, <label for="..."></code>	Groups the registration fields and ties each label to its corresponding input.
<code><input type="radio"> / type="checkbox"></code>	Let the user pick one gender option or multiple hobby options.
<code><select>, <option></code>	Provides a dropdown list for choosing the student's branch.
<code><textarea></code>	Accepts multi-line free-text comments or queries.
<code><header>, <nav>, <section>, <article>, <footer></code>	Give semantic meaning to the banner, navigation, content blocks, and page footer.
<code><audio controls>, <video controls poster="..."></code>	Embed playable audio and video with built-in browser controls.
<code><iframe src="..."></code>	Embeds an external YouTube video player inside the page.
<code><input type="date">/"number">/"range"></code>	Provide a calendar picker, a stepper, and a draggable slider without extra scripting.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load index.html in a browser	Banner, headings, and both list types render correctly	Page loaded with all Part A content visible	Pass
TC-02	Inspect the "Text Formatting Examples" paragraph	Bold, italic, underline, subscript, and superscript display correctly	All five formatting styles rendered as expected	Pass
TC-03	Inspect the Student Details table	Table shows a header row and three data rows in a bordered grid	Table rendered correctly with all sample records	Pass
TC-04	Fill the registration form and select a gender, hobbies, and branch	All entered values are reflected in their respective fields	All fields, radio buttons, checkboxes, and dropdown updated correctly	Pass
TC-05	Leave the required Name or Email field blank and click Register	Browser blocks submission and highlights the missing required field	Browser displayed its built-in "required" validation prompt	Pass
TC-06	Click the Clear Form (reset) button after filling the form	All fields return to their empty/default state	Form fields cleared successfully	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-07	Enter a value above 30 or below 17 in the Age field	Number input restricts or flags the out-of-range value	Browser enforced the min="17"/max="30" constraint	Pass
TC-08	Drag the Skill Proficiency range slider	Slider value updates smoothly between 0 and 10	Slider moved and updated correctly	Pass

9. OUTPUT

Output 1: Initial Page Load — Part A and Part B

The page renders the banner, the About ANITS paragraph with mixed text formatting, the numbered and bulleted lists, the hyperlinks, the bordered Student Details table, and the two campus images.

College Information Portal

Part A - Basic HTML

About ANITS

Anil Neerukonda Institute of Technology and Sciences (ANITS) is a premier engineering institution that provides **quality education** and actively encourages *innovation*, research, and practical engineering skills.

Text Formatting Examples

Bold Text | *Italic Text* | Underlined Text

Chemical formula: H₂O

Mathematical expression: $x^2 + y^2$

Courses Offered

1. Computer Science & Engineering
2. Information Technology
3. CSE (AI & ML)
4. Electronics & Communication Engg.

Campus Facilities

- Central Library & e-Learning Center
- High-Performance Computing Labs
- In-Campus Hostel Facility
- Sports & Athletic Grounds

Useful Websites

[ANITS Official Website](#) | [Google Search](#)

Part B - Tables and Images

Student Details

Roll No	Name	Branch	Email
A24126552093	GANESH	CSE(AIML)	GANESH@gmail.com
A24126552092	TARUN	CSE	TARUN@gmail.com
A24126552094	PRABHAT	IT	PRABATH@gmail.com

College Brand & Campus

Official Logo:

 ANITS College Logo

Campus Building (Admin Block):

 ANITS Campus

Output 2: Filled Registration Form — Part C

The registration form is filled with sample student data — full name, email, password, gender, hobbies, branch, and comments — showing the radio buttons, checkboxes, dropdown, and textarea in use.

Part C - Student Registration Form

Full Name:

Email Address:

Password:

Gender:
☒ Male ☐ Female ☐ Other

Hobbies:
☒ Reading ☐ Music ☒ Coding

Select Branch:

Comments / Queries:

Requesting hostel accommodation for the upcoming semester.

Output 3: HTML5 Multimedia and Modern Inputs — Part D

The video player, along with the HTML5 email, date, number, and range controls, renders with its native browser UI, including the calendar picker, numeric stepper, and draggable slider.

HTML5 Video Player:



HTML5 Interactive Controls

Email Control:

Joining Date: 

Age (17 - 30): 

Skill Proficiency (0 - 10): 

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10. RESULT

Thus, a College Information Portal web page was successfully designed and developed using HTML text formatting tags, ordered and unordered lists, hyperlinks, tables, images, a multi-field registration form, and HTML5 semantic and multimedia elements. The page demonstrates structured content presentation, a functional registration form with varied input types, and the use of modern HTML5 controls. All eight test cases passed successfully, confirming correct rendering, form behaviour, and input validation.